BASIC SKILLS BOOKLET

Algebra Basic Skills

Learn it. Practise it. Show you can do it.

Name: _____ Class: _____

My progress

	Skill	Practice done	Skill Check
1	**Collecting Like Terms** I can add and subtract like terms, e.g. $3a + 4a = 7a$	☐	/ 4
2	**Multiplying Algebra** I can multiply terms, e.g. $2a \times 3b = 6ab$ and $x \times x = x^2$	☐	/ 4
3	**Dividing Algebra** I can divide terms, e.g. $8x \div 2 = 4x$ and $6ab \div 2a = 3b$	☐	/ 4
4	**Expanding Brackets** I can expand brackets, e.g. $3(x+2) = 3x+6$ and $-2(x-4) = -2x+8$	☐	/ 4
5	**Factorising into Brackets** I can factorise, e.g. $2x + 6 = 2(x + 3)$	☐	/ 4
6	**Substitution** I can substitute numbers into expressions, e.g. if $a = 4$ then $3a = 12$	☐	/ 4
		Skill Check total	/ 24

1 Collecting Like Terms

LEARN

Goal: I can add and subtract like terms, e.g. $3a + 4a = 7a$

Like terms have **exactly the same letter part**. You can only add or subtract like terms.

Like terms ✓	NOT like terms ✗
$3a$ and $5a$	$3a$ and $5b$ (different letters)
x and $7x$	x and 7 (one has no letter)
$2x^2$ and x^2	x^2 and x (x^2 is not the same as x)
$4ab$ and ab	ab and a

Think of the letters as objects. 3 apples + 4 apples = 7 apples, so $3a + 4a = 7a$. But 3 apples + 2 bananas can't be joined, so $3a + 2b$ stays as $3a + 2b$.

A letter on its own means 1 of it. a is the same as $1a$. So $a + a + a = 3a$.

Worked examples

$3a + 4a$

3 lots of a plus 4 lots of a

$= \mathbf{7a}$

$6x - 2x$

6 take away 2 is 4

$= \mathbf{4x}$

$5y - y$

y means $1y$, and $5 - 1 = 4$

$= \mathbf{4y}$

$2a + 3b + 4a + b$

The a's: $2a + 4a = 6a$

The b's: $3b + b = 4b$

$= \mathbf{6a + 4b}$

$x^2 + 3x + 2x^2$

The x^2's: $x^2 + 2x^2 = 3x^2$

The x's: just $3x$

$= \mathbf{3x^2 + 3x}$

$7m + 2 - 3m$

The m's: $7m - 3m = 4m$

The numbers: just 2

$= \mathbf{4m + 2}$

Remember: The sign ($+$ or $-$) in **front** of a term belongs to that term. Circle or highlight each type of term in a different colour before you add them up.

1 Collecting Like Terms

PRACTICE

Simplify by collecting like terms. Show your working in each box.

1 $a + a + a$

$=$ _____

2 $2a + 3a$

$=$ _____

3 $7b - 3b$

$=$ _____

4 $6y - y$

$=$ _____

5 $2p + 3p + p$

$=$ _____

6 $3x^2 + 4x^2$

$=$ _____

7 $2a + 3b + 4a$

$=$ _____

8 $5x + 2y + x$

$=$ _____

9 $4x + 3 + 2x$

$=$ _____

10 $5a + 2b - 3a$

$=$ _____

11 $x^2 + 2x + 3x$

$=$ _____

12 $2x^2 + x + x^2$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong.**

2 Multiplying Algebra

LEARN

Goal: I can multiply terms, e.g. $2a \times 3b = 6ab$ and $x \times x = x^2$

In algebra we **don't write the $\times$ sign**. Letters and numbers that are multiplied are written next to each other, **number first**.

We write...	...it means
$3a$	$3 \times a$
ab	$a \times b$
$5xy$	$5 \times x \times y$
x^2	$x \times x$ (say "x squared")

How to multiply terms

1. **Multiply the numbers** together.
2. **Write the letters** after the number, in alphabetical order.
3. If the **same letter** appears twice, write it as a **square**: $x \times x = x^2$.

Worked examples

$4 \times a$

Number first, then the letter

$= \mathbf{4a}$

$a \times b$

Write the letters together

$= \mathbf{ab}$

$2a \times 4$

Numbers: $2 \times 4 = 8$

Letters: a

$= \mathbf{8a}$

$2a \times 3b$

Numbers: $2 \times 3 = 6$

Letters: a and b

$= \mathbf{6ab}$

$x \times x$

The same letter twice is "squared"

$= \mathbf{x^2}$

$3x \times 2x$

Numbers: $3 \times 2 = 6$

Letters: $x \times x = x^2$

$= \mathbf{6x^2}$

Remember: When you **multiply**, the numbers multiply ($3 \times 2 = 6$) — don't add them! And $x \times x$ is x^2, not $2x$.

Going further: $x \times x \times x = x^3$ ("x cubed"), and $x^2 \times x = x^3$.

2 Multiplying Algebra

PRACTICE

Simplify. Show your working in each box.

1 $4 \times a$

$=$ _____

2 $a \times b$

$=$ _____

3 $2 \times 3a$

$=$ _____

4 $3x \times 4$

$=$ _____

5 $a \times a$

$=$ _____

6 $2a \times b$

$=$ _____

7 $2a \times 3b$

$=$ _____

8 $4x \times 2y$

$=$ _____

9 $3a \times a$

$=$ _____

10 $2x \times 5x$

$=$ _____

11 $4m \times 3m$

$=$ _____

12 $x \times x \times x$

$=$ _____

✓ Check your answers at the back. Fix any you got wrong.

3 Dividing Algebra

LEARN

Goal: I can divide terms, e.g. $8x \div 2 = 4x$ and $6ab \div 2a = 3b$

Dividing is the opposite of multiplying. A division can also be written as a fraction: $6a \div 2$ is the same as $\frac{6a}{2}$.

How to divide terms

1. **Divide the numbers.**
2. **Cancel any letter** that is on the top **and** the bottom.
3. Write down what is left.

Why can letters cancel?. Anything divided by itself is 1. Just like $5 \div 5 = 1$, $a \div a = 1$. So $\frac{ab}{a} = b$.

Worked examples

$6a \div 2$

Numbers: $6 \div 2 = 3$

Letters: a stays

$= \boldsymbol{3a}$

$ab \div a$

The a cancels

b is left

$= \boldsymbol{b}$

$8x \div x$

The x cancels

8 is left

$= \boldsymbol{8}$

$12ab \div 4a$

Numbers: $12 \div 4 = 3$

a cancels, b is left

$= \boldsymbol{3b}$

$x^2 \div x$

x^2 is $x \times x$

One x cancels, one is left

$= \boldsymbol{x}$

$6x^2 \div 3x$

Numbers: $6 \div 3 = 2$

Letters: $x^2 \div x = x$

$= \boldsymbol{2x}$

Remember: If **all** the letters cancel, you are left with just a number, e.g. $5a \div a = 5$. Write x^2 as $x \times x$ if it helps you see what cancels.

3 Dividing Algebra

PRACTICE

Simplify. Show your working in each box.

1 $8a \div 2$

$=$ _____

2 $12x \div 3$

$=$ _____

3 $20y \div 4$

$=$ _____

4 $6m \div 6$

$=$ _____

5 $ab \div b$

$=$ _____

6 $5a \div a$

$=$ _____

7 $6ab \div 2a$

$=$ _____

8 $12xy \div 4y$

$=$ _____

9 $x^2 \div x$

$=$ _____

10 $4x^2 \div x$

$=$ _____

11 $6x^2 \div 2x$

$=$ _____

12 $15a^2 \div 5a$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong.**

4 Expanding Brackets

LEARN

Goal: I can expand brackets, e.g. $3(x+2) = 3x+6$ and $-2(x-4) = -2x+8$

Expanding means getting rid of the brackets by multiplying. A number or letter right in front of a bracket means **multiply everything inside** the bracket by it.

Multiply the outside by EVERY term inside
Draw arrows so you don't miss one:

$$3(x+2) \longrightarrow 3\times x \text{ and } 3\times 2 \longrightarrow 3x+6$$

Worked examples

$4(a+5)$

$4\times a = 4a$
$4\times 5 = 20$
$= \mathbf{4a+20}$

$2(x-3)$

$2\times x = 2x$
$2\times(-3) = -6$
$= \mathbf{2x-6}$

$x(x+5)$

$x\times x = x^2$
$x\times 5 = 5x$
$= \mathbf{x^2+5x}$

Negative numbers outside the bracket

When the number outside is **negative**, use the sign rules:

Signs	Answer
same signs: $+\times+$ or $-\times-$	**positive** $(+)$
different signs: $+\times-$ or $-\times+$	**negative** $(-)$

$-2(x+3)$

$-2\times x = -2x$
$-2\times 3 = -6$
$= \mathbf{-2x-6}$

$-3(x-4)$

$-3\times x = -3x$
$-3\times(-4) = +12$
$= \mathbf{-3x+12}$

$-(x-5)$

A minus on its own means -1
$-1\times x = -x,\ -1\times(-5) = +5$
$= \mathbf{-x+5}$

Remember: The outside term multiplies **every** term inside: $3(x+2)$ is $3x+6$, **not** $3x+2$. With a negative outside, **every sign inside changes**: $-2(x-4) = -2x+8$.

4 Expanding Brackets

PRACTICE

Expand. Be careful with the negative signs! Show your working in each box.

1 $2(x+3)$

$=$ _____

2 $3(a+4)$

$=$ _____

3 $4(y-2)$

$=$ _____

4 $3(2x+1)$

$=$ _____

5 $x(x+2)$

$=$ _____

6 $x(x-3)$

$=$ _____

7 $-2(x+3)$

$=$ _____

8 $-4(x-2)$

$=$ _____

9 $-(x+7)$

$=$ _____

10 $2(3-x)$

$=$ _____

11 $-2(3x-1)$

$=$ _____

12 $-x(x+2)$

$=$ _____

✓ Check your answers at the back. Fix any you got wrong.

5 Factorising into Brackets

LEARN

Goal: I can factorise, e.g. $2x + 6 = 2(x + 3)$

Factorising is the **opposite of expanding**. We put the brackets back in.

$$2(x+3) \xrightarrow{\text{expand}} 2x+6 \qquad 2x+6 \xrightarrow{\text{factorise}} 2(x+3)$$

How to factorise

1. Find the **biggest number** (and/or **letter**) that goes into **every** term. Write it outside the bracket.
2. **Divide** each term by it. Write the answers inside the bracket.
3. **Check** by expanding — you should get back to the question.

Worked examples

$5a + 10$

5 goes into both

$5a \div 5 = a,\ 10 \div 5 = 2$

$= \mathbf{5(a + 2)}$

$6x - 9$

3 goes into both

$6x \div 3 = 2x,\ -9 \div 3 = -3$

$= \mathbf{3(2x - 3)}$

$x^2 + 4x$

x goes into both

$x^2 \div x = x,\ 4x \div x = 4$

$= \mathbf{x(x + 4)}$

$3x^2 + 6x$

$3x$ goes into both

$3x^2 \div 3x = x,\ 6x \div 3x = 2$

$= \mathbf{3x(x + 2)}$

$14 - 7x$

7 goes into both

$14 \div 7 = 2,\ -7x \div 7 = -x$

$= \mathbf{7(2 - x)}$

$8x - 12$

4 is the **biggest**

(2 works, but isn't the biggest)

$= \mathbf{4(2x - 3)}$

Remember: Keep the sign with each term: in $6x - 9$ the second term is -9, so the bracket has a minus: $3(2x - 3)$. Always **check by expanding** your answer.

5 Factorising into Brackets

PRACTICE

Factorise fully. Show your working in each box.

1 $3x + 6$

$=$ _____

2 $2a + 8$

$=$ _____

3 $4y - 12$

$=$ _____

4 $6a + 9$

$=$ _____

5 $x^2 + 3x$

$=$ _____

6 $x^2 - 2x$

$=$ _____

7 $8x - 12$

$=$ _____

8 $6a - 15$

$=$ _____

9 $14 - 7x$

$=$ _____

10 $ab + 3a$

$=$ _____

11 $3x^2 + 6x$

$=$ _____

12 $2a^2 - 8a$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong.**

6 Substitution

LEARN

Goal: I can substitute numbers into expressions, e.g. if $a = 4$ then $3a = 12$

Substitution means **replacing each letter with a number**, then working out the answer.

How to substitute

1. Write the expression out again.
2. **Replace each letter** with its number. Put in the hidden $\times$ signs.
3. **Work it out.** Do $\times$ and $\div$ before $+$ and $-$ (BEDMAS).

Watch out for the hidden $\times$. $3a$ means $3 \times a$. So if $a = 4$, then $3a = 3 \times 4 = 12$ — **not** 34!
a^2 means $a \times a$. So if $a = 3$, then $a^2 = 3 \times 3 = 9$ — **not** 6!

Worked examples

$a + 5$
when $a = 3$

Replace a with 3
$3 + 5$
$= 8$

$4b$
when $b = 2$

$4b$ means $4 \times b$
4×2
$= 8$

$2x + 1$
when $x = 5$

$2 \times 5 + 1$
$= 10 + 1$
$= 11$

ab
when $a = 3,\ b = 4$

ab means $a \times b$
3×4
$= 12$

x^2
when $x = 6$

x^2 means $x \times x$
6×6
$= 36$

$3a - b$
when $a = 4,\ b = 2$

$3 \times 4 - 2$
$= 12 - 2$
$= 10$

Remember: Put a **negative** number in brackets when you substitute it: if $x = -3$, then $2x = 2 \times (-3) = -6$.

6 Substitution

PRACTICE

Find the value of each expression. Show your working in each box.

1 $x + 7$ when $x = 3$

$=$ _____

2 $a - 2$ when $a = 9$

$=$ _____

3 $5y$ when $y = 4$

$=$ _____

4 $3m$ when $m = 6$

$=$ _____

5 $2x + 3$ when $x = 4$

$=$ _____

6 $4a - 5$ when $a = 3$

$=$ _____

7 ab when $a = 2,\ b = 5$

$=$ _____

8 x^2 when $x = 5$

$=$ _____

9 $3a + 2b$ when $a = 2,\ b = 3$

$=$ _____

10 $a^2 + 1$ when $a = 4$

$=$ _____

11 $2x + 5$ when $x = -1$

$=$ _____

12 $4 - 3x$ when $x = -2$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong.**

Skill Checks

TEACHER MARKS

Name: _____

Do these **on your own — no notes**. Show your working, and write your final answer on the line.

1 Collecting Like Terms

Simplify by collecting like terms.

Mark: ____ / 4

1 $5a + 2a$

= _____

2 $8x - 3x$

= _____

3 $4a + 3b + 2a$

= _____

4 $x^2 + 5x + 2x^2$

= _____

2 Multiplying Algebra

Simplify.

Mark: ____ / 4

1 $3 \times 5a$

= _____

2 $2a \times 4b$

= _____

3 $x \times x$

= _____

4 $3x \times 2x$

= _____

3 Dividing Algebra

Simplify.

Mark: ____ / 4

1 $12x \div 4$

= _____

2 $ab \div b$

= _____

3 $10ab \div 5a$

= _____

4 $8x^2 \div 2x$

= _____

Skill Checks

CONTINUED

Name: _____

Total: ____ / 24

Do these **on your own — no notes**. Show your working, and write your final answer on the line.

4 Expanding Brackets

Expand.

Mark: ____ / 4

1

$3(x + 5)$

$=$ _____

2

$2(4a - 3)$

$=$ _____

3

$x(x - 2)$

$=$ _____

4

$-4(x - 3)$

$=$ _____

5 Factorising into Brackets

Factorise fully.

Mark: ____ / 4

1

$5x + 20$

$=$ _____

2

$6a - 9$

$=$ _____

3

$x^2 + 8x$

$=$ _____

4

$4y^2 - 12y$

$=$ _____

6 Substitution

Find the value.

Mark: ____ / 4

1

$4a$

when $a = 5$

$=$ _____

2

$3x + 2$

when $x = 4$

$=$ _____

3

xy

when $x = 3,\ y = 6$

$=$ _____

4

$y^2 - 3$

when $y = 5$

$=$ _____

Answers to the Practice Questions

Terms can be in a different order and still be right — e.g. $2b + 3a$ is the same as $3a + 2b$.

1. Collecting Like Terms

1. $3a$
2. $5a$
3. $4b$
4. $5y$
5. $6p$
6. $7x^2$
7. $6a + 3b$
8. $6x + 2y$
9. $6x + 3$
10. $2a + 2b$
11. $x^2 + 5x$
12. $3x^2 + x$

2. Multiplying Algebra

1. $4a$
2. ab
3. $6a$
4. $12x$
5. a^2
6. $2ab$
7. $6ab$
8. $8xy$
9. $3a^2$
10. $10x^2$
11. $12m^2$
12. x^3

3. Dividing Algebra

1. $4a$
2. $4x$
3. $5y$
4. m
5. a
6. 5
7. $3b$
8. $3x$
9. x
10. $4x$
11. $3x$
12. $3a$

4. Expanding Brackets

1. $2x + 6$
2. $3a + 12$
3. $4y - 8$
4. $6x + 3$
5. $x^2 + 2x$
6. $x^2 - 3x$
7. $-2x - 6$
8. $-4x + 8$
9. $-x - 7$
10. $6 - 2x$
11. $-6x + 2$
12. $-x^2 - 2x$

5. Factorising into Brackets

1. $3(x + 2)$
2. $2(a + 4)$
3. $4(y - 3)$
4. $3(2a + 3)$
5. $x(x + 3)$
6. $x(x - 2)$
7. $4(2x - 3)$
8. $3(2a - 5)$
9. $7(2 - x)$
10. $a(b + 3)$
11. $3x(x + 2)$
12. $2a(a - 4)$

6. Substitution

1. 10
2. 7
3. 20
4. 18
5. 11
6. 7
7. 10
8. 25
9. 12
10. 17
11. 3
12. 10